Jehanzeb Ali

Applied AI engineer: agents, evals and guardrails · game AI · native Urdu.

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SUMMARY

I build AI agents that act in 3D worlds, plus the evals and guardrails that prove what they did. I contribute upstream to Autopilot, an open-source agent that plays and tests games; I measure how prompt-injection defences fail in Urdu and Roman Urdu; and I founded Valar, an AI traffic firewall. Behind that are 5+ years of shipping real-time Roblox and Unity VR systems to large player bases.

AI WORK

- **Worldsmith** (Nov 2026 – present, repo). An AI architect agent repairs a broken 3D game *world* so the same frozen player can complete it. Rooms are built in Roblox Studio over MCP from a bounded JSON grammar, and a validator rejects patches that cheat (69 abuse cases rejected, 13 legitimate patches accepted in its test suite). First live result: on a room where the scripted player fell 3 of 3 times, Claude Haiku, Sonnet and Opus each proposed one patch that passed the validator, and the rebuilt room was completed 3 of 3 (one episode per model, $0.04 of API calls). Next: harder rooms, equal-budget baselines and a local Gemma model.
- **Autopilot upstream contributions** (ramenvr/autopilot, Apache-2.0, an AI agent that plays and tests games). 3 PRs merged into its `staging` branch on 7 Nov 2026: #40 `roblox-mcp` (plugin-free Roblox Studio connector), #38 `autopilot regress` (model-free replay of confirmed results) and #39 `evolve` (autoresearch loop over the prompt, kit and constants).
- **Lost in Transliteration** (prompt-injection research in Urdu and Roman Urdu, repo). Meta Prompt Guard 2 (86M) catches 48.5% of English injection attacks but only 11.5% of the same attacks in Roman Urdu (194/400 vs 46/399). Evaluated open-source injection classifiers, 5 LLMs on 927 tasks each (Claude Haiku, Sonnet and Opus 5.5, GPT-5.4 nano, Gemma 4 31B) and MCP security scanners; fine-tuned small multilingual detectors (XLM-R, mDistilBERT) to close the gap.
- **Agent firewall** (action-level guardrail for tool-using agents, repo). On attack styles held out of training, injection success against scripted agents fell from 23.3% to 0.2% (8 of 5,000 attacked episodes) while blocking 0.9% of legitimate actions; wallet drains fell from 97.9% to 0% across 1,102 malicious requests. Synthetic attacks; next is real Claude agents and tasks it hasn't seen.
- **Valar** (founder, Apr 2026 – present). OpenAI-compatible AI traffic firewall between a company's staff, apps and agents and its LLM providers: detects secrets, personal data and prompt injection in English, Urdu and Roman Urdu, with a policy engine (observe, alert, redact, block, require approval) and metadata-only audit logs.

Local AI lab and agent tooling (NVIDIA DGX Spark, 2025 – present)

- Run a personal lab on an NVIDIA DGX Spark (Grace Blackwell, 128 GB unified memory) to fine-tune, evaluate and serve open-weight LLMs locally, the same setup behind Valar's on-prem option.
- Fine-tune open-weight models with LoRA and QLoRA (PyTorch, Hugging Face Transformers, PEFT, TRL, Unsloth) and serve them with vLLM, including Gemma 4 31B as a local teacher model.
- Built JzLab, an orchestrator that starts and steers Claude Code agents on the Spark over SSH; spark-lab, a self-healing GPU job queue; spark-one, a calibrated small decision model; and Scam Radar, an on-device scam checker.

EXPERIENCE

TableSide | Founder & CEO · Jul 2026 – Present · Karachi, Pakistan

- Built a fully automated photo-to-3D pipeline that turns phone photos of dishes into 3D models for AR menus and marketing banners, at about 10 minutes per dish with no manual input.
- Launched Asia's first AR menus with 5+ restaurant brands: 130+ dishes modelled and 4,000 diner scans in 4 weeks.
- Architected for low marginal cost (Cloudflare R2/CDN, Fly.io, Supabase, Vercel) at under $20 per menu per month; shipped contactless ordering with POS integration and priority-based review alerting.

Voldex | Software Engineer · Feb 2025 – Apr 2026 · Remote

- Engineered and optimised live Roblox experiences in Luau for a large, active player base, reducing memory use and improving responsiveness.
- Shipped monetisation systems to production on schedule with design and art teams, increasing engagement and revenue.

Remio VR | Unity Developer · Nov 2020 – Feb 2025 · Remote

- Built large-scale VR environments in Unity for titles supporting 100k+ monthly active users, and cut loading times by 25% through refactoring and resource optimisation.
- Delivered monthly releases for four years across time zones (Jira, Linear).

Selected achievements: worked on NFL Football Universe, a live Roblox experience with 400k+ daily active users; published 15+ VR experiences to the Meta Store.

SKILLS

- **AI engineering:** Python, TypeScript, PyTorch, Hugging Face (Transformers, PEFT, TRL), vLLM, Unsloth/LoRA/QLoRA, MCP, Claude Agent SDK, LLM APIs (Anthropic, OpenAI, Google), RAG, agentic workflows and tool use, quantisation and local inference.
- **Evals and safety:** eval-set design, held-out splits and baselines, calibration, red-teaming, prompt-injection detection, guardrails, PII redaction and DLP, audit logging.
- **Games and real-time:** Luau, C#, C++, Roblox Studio, Unity (VR).
- **Cloud:** Cloudflare R2/CDN, Fly.io, Supabase, Vercel.
- **Product:** product strategy, go-to-market, pilot sales, remote team coordination, Agile/Scrum.

LANGUAGES

Urdu and Roman Urdu (native), English (fluent).

EDUCATION

Staffordshire University | BSc (Hons) Games Design & Programming · Second Class Honours · Graduated June 2020